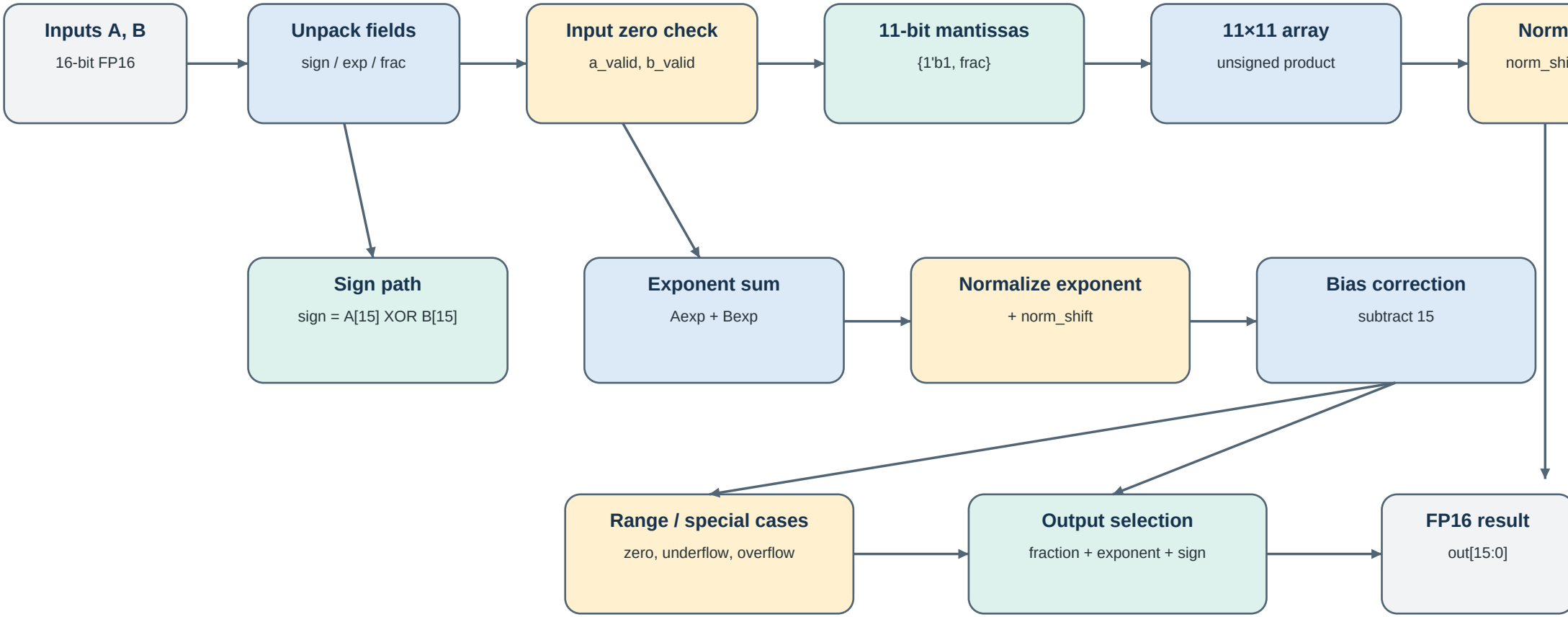
FP16 Multiplier — Top-Level Data Path

Block-level reconstruction from fp16_multiplier_core.v and array_mul_11bit.v



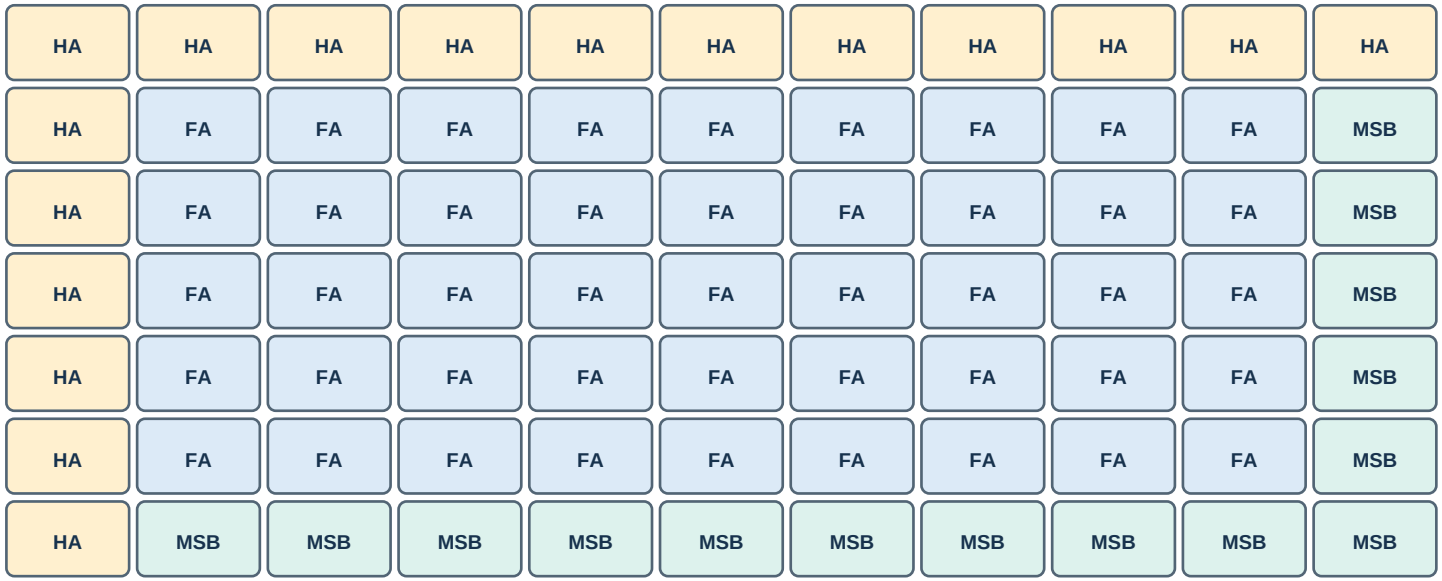
Implementation rule: all non-primitive logic is built hierarchically from c1/c2/s1/s2-based modules.

Project special cases and truncation behavior are retained; no rounding stage is present.

11×11 Array Multiplier — Internal Structure

The mantissas include the implicit leading 1, so both operands are unsigned 11-bit values.

Rows correspond to bits of a; columns correspond to bits of b.



Schematic view (illustrative subset of rows; actual array is 11×11).

Generic inner cell

partial product + sum + carry

Generic cell: p = x AND y; sum = p XOR incoming_partial XOR carry. Carry is selected with a c1 mux, avoiding a separate full-adder structure.

Boundary cell

one input is constant zero

Top row and left column specialize the half-adder behavior. The corner has no incoming partial product or carry.

Leading-bit cell

a[10]=1 or b[10]=1

The implicit leading one makes the partial product equal to the other operand bit, so an AND stage is removed.

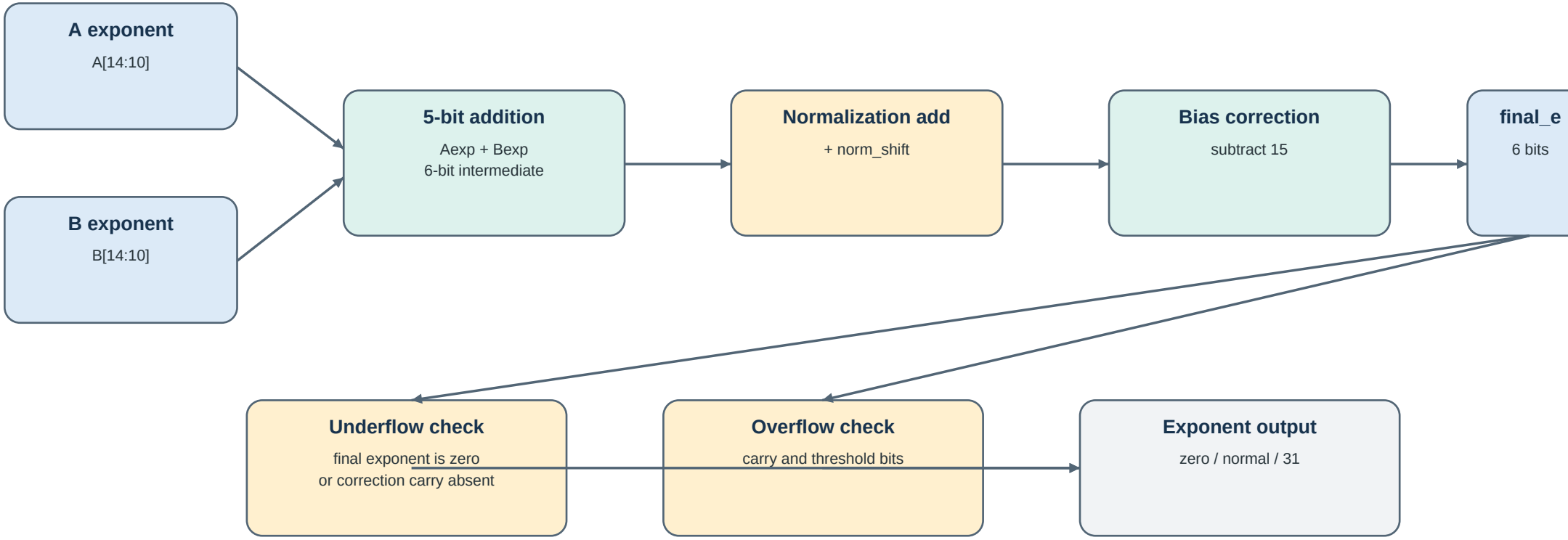
Array outputs

22-bit product P[21:0]

The array emits the complete 22-bit product. The top product bit controls whether the fraction is shifted during normalization.

Exponent Path — Sum, Normalize, Bias Correction

The exponent logic is separate from the 11×11 mantissa multiplication.



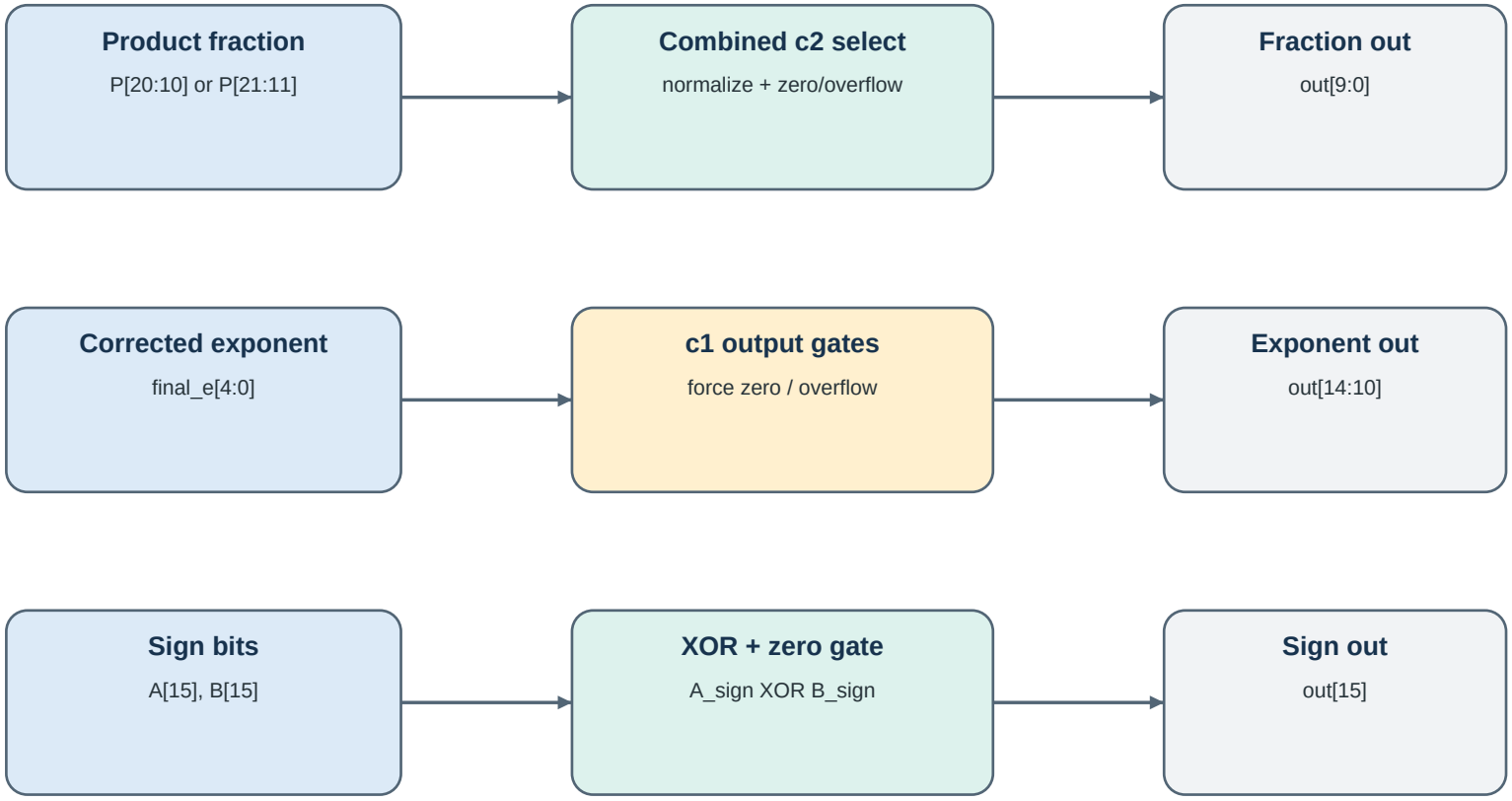
The first stage adds the two 5-bit stored exponents and preserves the carry as bit 5. The next stage increments the intermediate exponent when the mantissa product requires a one-bit normalization shift.

The bias correction is implemented as subtraction of the FP16 bias (15). The result is checked for underflow and overflow before selecting the final 5-bit exponent.

As required by the project, overflow returns exponent 31 and zero fraction; underflow or a zero input forces the output to zero. The datapath truncates rather than rounds.

Output Logic and Structural Cell Choices

Selection logic is merged where possible to reduce repeated mux/cell instances.



Cell costs

c1 = 10 gates
c2 = 11 gates
s1 = 13 gates
s2 = 13 gates

The fraction output cell selects the normalized slice and forces zero when zero/overflow conditions apply. This removes a separate fraction mux layer.

Structural strategy

Use c1/c2/s1/s2 only
for composed logic.
No behavioral arithmetic operators
in the hand-built datapath.

The sign is the XOR of input signs, except that a zero input forces the final sign to zero. The exponent output is forced to all ones for overflow.

Note: diagrams are explanatory block schematics, not transistor-level drawings. Exact gate total must be confirmed by the provided simulation flow.